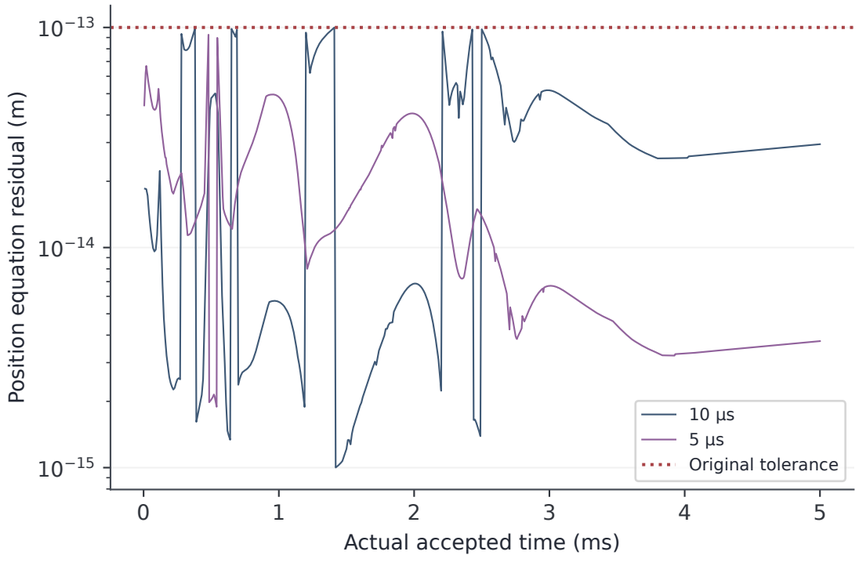
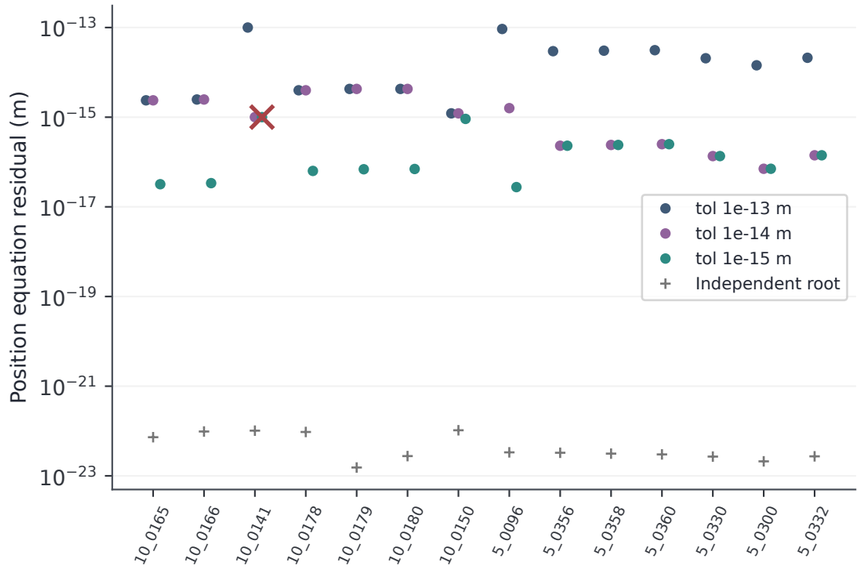
B2 solver tolerance and time discretization: distinct numerical tests

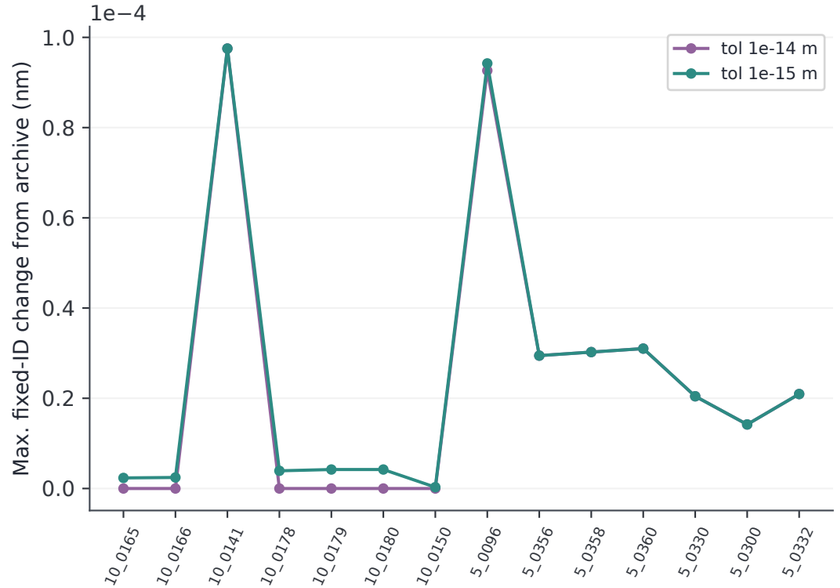
a Archived nonlinear residuals



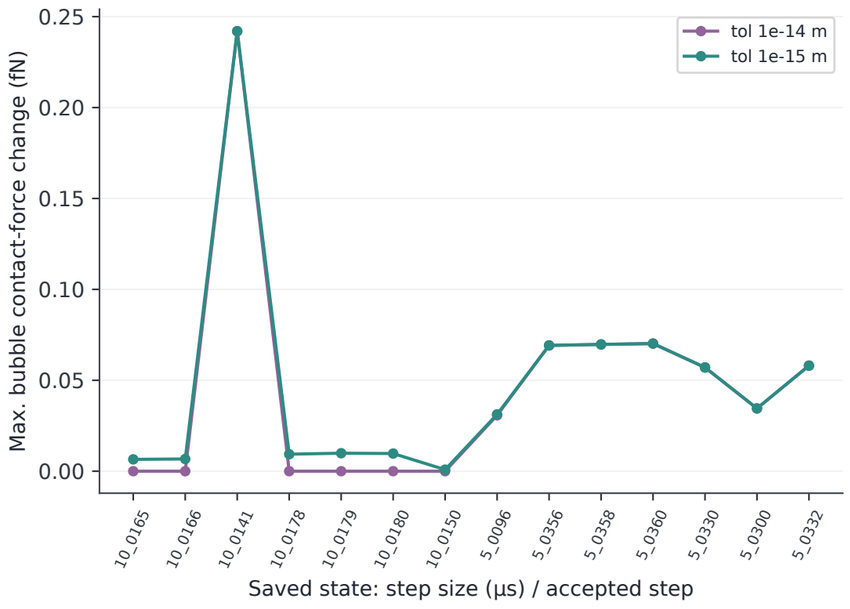
b 14 same-input CPU reference states



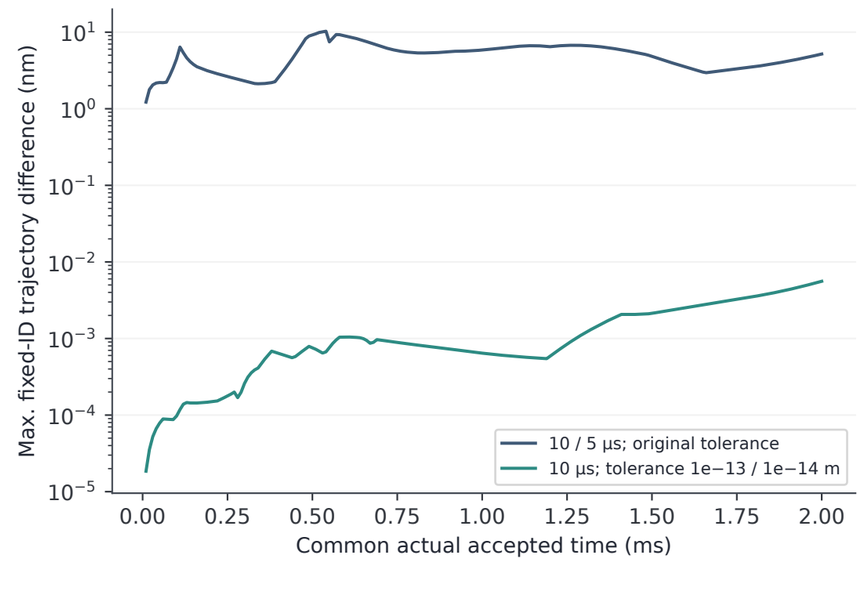
c Position changes after tightening



d Active contacts and edges unchanged



e One original-initial-state 2 ms control



f Limits and observed event sensitivity

1e−14 m tolerance: 14/14 states converge
1e−15 m: 13/14 (red × marks failure)
Failed residual: 1.0026e−15 m

2 ms tighter-tolerance replay:
200 accepted steps; max. ID change 0.00559 nm
All 23 prefix event times unchanged
Worst 10 / 5 µs edge lag still 130 µs

Near-threshold slope amplifies small gap changes.
2.5 µs reference: NOT_RUN
No absolute temporal-order claim.

Same stiffness, acoustic force, CFD and discrete resistance law; no additional contact model. Nonconverged CPU iterate is retained as a failure.
Tighter local solves alone do not bound trajectory error; the single 2 ms replay tests propagation without an unverified checkpoint restart.